MW-AeroFusion-Net: Fusing Satellite, Reanalysis and Ground-Station Data for Leakage-Safe $PM_{2.5}$ Forecasting in Indian Cities

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***Abstract*—Station-level forecasts of fine particulate matter ($PM_{2.5}$) one to two days ahead are needed for public health advisories in Indian cities, yet learned forecasting models are rarely compared with an operational chemistry-transport forecast under inputs that would actually be available at issue time. We present a leakage-safe benchmark over a fixed panel of 116 Central Pollution Control Board (CPCB) stations in 28 Indian cities, in which every input respects its real publication latency and the test years are opened once, after the model is frozen. On this benchmark we evaluate MW-AeroFusion-Net, a five-component model that combines causal wavelet bands, a wind-aware adaptive diffusion graph over stations and upwind source regions, a ConvLSTM over gridded meteorology, fire and CAMS fields, a Transformer temporal encoder, and source-group attribution gating. We compare it with persistence, climatology, raw and bias-corrected CAMS global forecasts, LightGBM and five neural baselines at +1, +24 and +48 h, in a past-only and a CAMS-informed setting. [TO FILL: one or two sentences of headline results from tables/test_sa_station.tex and test_sb_station.tex: RMSE and skill versus persistence and versus CAMS with bias correction at +24 and +48 h, with Diebold–Mariano significance]. A component ladder and leave-one-out ablations show [TO FILL: which components helped and which did not, from the ladder and ablation tables]. The learned attribution of the gating layer [TO FILL: passed / failed] faithfulness, randomisation and plausibility checks. Code, configuration and the station panel are released.**

***Index Terms*—$PM_{2.5}$ forecasting, graph neural networks, CAMS, air quality, India, wavelet, ConvLSTM, explainability, benchmark**

I. Introduction

Fine particulate matter ($PM_{2.5}$) is the dominant air-quality risk in Indian cities. Winter inversions, crop-residue burning in Punjab and Haryana, festival fireworks and pre-monsoon dust produce episodes in which concentrations exceed the national standard many times over. Operational forecasts in India come from chemistry-transport systems such as SAFAR [1] and the IITM early-warning system [2], [3], and from global products such as the Copernicus Atmosphere Monitoring Service (CAMS) forecasts. In parallel, learned spatio-temporal models for $PM_{2.5}$ are well established [4], [5], [6], [7], including graph models for Indian stations [8], [9], and wavelet or decomposition hybrids are common [10], [11], [12], [13].

Two questions are rarely answered in this literature. First, does a learned model beat a calibrated CAMS forecast at matched lead times when its inputs are restricted to what was published at issue time? Many studies feed reanalysis meteorology for future hours [4], normalise or decompose the whole series before splitting [14], or compare only against other neural networks. Second, which parts of a composite architecture actually help? Multi-component models are usually reported as a whole, so the contribution of each part is unknown.

We address both questions with a benchmark and a component analysis. Our contributions are:

1) A *leakage-safe protocol* for station-level $PM_{2.5}$ forecasting over India: a fixed station panel, publication latencies for every input source, causal feature construction, training-only normalisation, and a test split that is read once by a single script after model freeze.
2) *MW-AeroFusion-Net*, which combines causal wavelet bands, a wind-aware adaptive diffusion graph, gridded ConvLSTM context, a Transformer encoder and source-group gating, built rung by rung so that each component's effect is measured, with leave-one-out ablations over five seeds.
3) A comparison against *CAMS global forecasts* at matched leads, raw and with per-station bias correction, together with LightGBM and five neural baselines, in a past-only setting and a CAMS-informed (post-processing) setting.
4) A *validation of the learned attribution* produced by the gating layer, using deletion tests, weight randomisation, seed stability and physical plausibility, reported whether each check passes or fails.

Code, configurations, the station panel and the split hashes are released.[1]

II. Related Work

Spatio-temporal graph models. Diffusion convolution [15] and learned adaptive adjacency [16] originated in traffic forecasting alongside spectral graph models [17], [18]. PM2.5-GNN [4] adds distance and wind-direction edges for $PM_{2.5}$ over Chinese cities, but its offline experiments feed reanalysis weather for future steps, which corresponds to our perfect-prognosis upper bound rather than an operational forecast. Attention-based models [19], [5], [6] and graph transformers [20], [21] report strong results on Chinese networks. For India, cluster-based and spatially attentive graph networks have been applied to CPCB stations [8], [9].

[1]Repository link withheld for double-blind review.

Wavelet and decomposition hybrids. Wavelet or mode decompositions followed by neural regressors are common [10], [11], [12], [13], [22]. Decomposing the full series before splitting leaks future samples into the inputs [14]; boundary-corrected transforms avoid this [23]. Several of the studies we checked do not state when the decomposition is applied relative to the split. We compute the transform causally inside each input window.

CAMS evaluation and post-processing. CAMS forecasts have been validated and calibrated with machine learning over China and the United States [24] and post-processed with model output statistics in Europe [25]. Over the Indo-Gangetic Plain, CAMS and its reanalysis [26] have been evaluated for aerosol composition and optical depth [27], [28], [29]. In our search we did not find a study that scores CAMS global $PM_{2.5}$ forecasts against CPCB stations at matched leads.

Interpretability. Variable-selection gating follows the Temporal Fusion Transformer [30]. Attention weights are not explanations by default [31], and saliency methods can fail randomisation tests [32]; we therefore test the gate weights with deletion, Integrated Gradients [33] and randomisation, and compare them with TreeSHAP [34] on LightGBM.

III. Data

Targets. Hourly CPCB $PM_{2.5}$ from the XKDR bulk export of the CPCB continuous monitoring network (CC BY 4.0). CPCB labels hours in Indian Standard Time at the start of the averaging interval; we convert them to UTC hours at ingest. Station counts in Indian cities grew several-fold over the study period, so city medians drift. We therefore fix a panel of 116 stations within 25 km of 28 of the 33 study cities: stations reporting since 2020-01-01 with at least 60% hourly completeness through 2024. The five cities without such a station stay in the graph as unlabelled nodes. Quality control removes sentinel and out-of-range values, flat-lines of six hours or more, and isolated spikes found with a trailing 24 h Hampel rule; a spike seen at two or more stations of the same city in the same hour is kept as a real event. Targets are never filled; missing targets are masked in the loss. Table I summarises the data card.

Inputs. (i) Pollutant history ($PM_{2.5}$, PM_{10}, NO_2, CO, O_3) with missing-value masks; gaps of up to three hours are filled forward, never interpolated. (ii) ERA5 meteorology [35] at each node: temperature, humidity, 10 m wind, pressure, precipitation, boundary-layer height and ventilation (wind speed $\times$ boundary-layer height). (iii) VIIRS and MODIS active fires from NASA FIRMS: counts and fire radiative power (FRP) within 300 km and upwind counts within 500 km. (iv) CAMS global forecasts of $PM_{2.5}$, total and dust aerosol optical depth. (v) Calendar and event flags: hour and day of year, weekday, public holidays, a Diwali window, the stubble-burning season and the 2020 lockdown. A gridded dataset on a 0.75° India grid (44×43 cells, 3-hourly) holds 10 m wind, boundary-layer height, temperature, rasterised FRP and CAMS fields for the ConvLSTM branch.

TABLE I
Data summary (train and validation years only; from DATA_CARD.md)

Item	Value
Stations in panel / cities with a station	116 / 28
Upwind context regions (no target)	17
Input window / forecast horizon	72 h / 48 h
$PM_{2.5}$ station-hours after QC (train+val)	[TO FILL: DATA_CARD]
$PM_{2.5}$ removed by QC, all rules (%)	[TO FILL: DATA_CARD]
Median city $PM_{2.5}$ coverage, train / val (%)	[TO FILL: DATA_CARD]

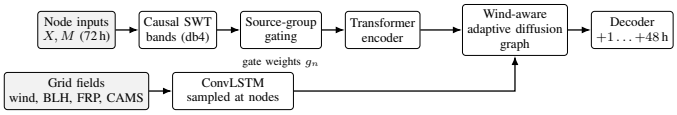


Fig. 1. MW-AeroFusion-Net. Per-node inputs are extended with causal wavelet bands, weighted by source-group gates and encoded over time; gridded context from a ConvLSTM is sampled at node locations; a diffusion graph mixes node states before a per-lead decoder. Rungs M0–M6 add these parts one at a time.

IV. Method

Fig. 1 shows the model. Let N be the number of nodes (stations, unlabelled cities and upwind regions), $T = 72$ the input window and $L = 48$ the horizon. The input at issue time t_0 is $X \in \mathbb{R}^{T \times N \times F}$ with mask M; the output is $\hat{y} \in \mathbb{R}^{L \times N}$, the standardised $\log(1 + PM_{2.5})$ at every lead.

Causal wavelet bands. For $PM_{2.5}$ and PM_{10} we compute an undecimated à-trous db4 transform with three levels inside each input window, using left-padded causal filters so that the band value at hour t depends only on samples at or before t. This avoids the leakage of full-series transforms [14]. The detail and approximation bands are appended as input channels.

Source-group attribution gating. Input channels are partitioned into seven groups: pollutant history, traffic proxy (NO_2, CO, weekday, holiday), meteorology, fire, CAMS, dust and calendar. For node n, a summary $s_{n,k}$ of group k (window mean and last value) gives gate weights

$$g_n = \mathrm{softmax}\left(W\,[\phi_1(s_{n,1}); \ldots; \phi_G(s_{n,G})]\right), \qquad (1)$$

and the per-step input is $h_{t,n} = \sum_k g_{n,k}\, P_k(x_{t,n,\mathcal{G}_k})$ with a linear projection P_k per group, in the spirit of TFT variable selection [30]. An entropy penalty $\lambda\, \mathbb{E}[-\sum_k g_{n,k} \log g_{n,k}]$ encourages sparse weights (an L1 penalty would be constant, since the weights sum to one). We call g_n a *learned attribution*; it is not a source apportionment.

Temporal encoder. A two-layer Transformer encoder [36] with a causal mask runs over the 72 steps of each node, and its last state is the node embedding e_n. Rungs M0–M2 use a GRU instead.

Regional context. A ConvLSTM [37] runs over the past grid fields, and its final hidden map is sampled bilinearly at each node's location and concatenated to e_n. Fire radiative power on the grid is hidden for the most recent hours to respect detection latency.

Wind-aware adaptive diffusion graph. Edges join each node to its eight nearest nodes within 600 km, with static weight $A_{ij} = \exp(-d_{ij}/\sigma)$ (σ the median neighbour distance), modulated at issue time by wind alignment

$$\tilde{A}_{ij}(t_0) = A_{ij}\ \max\big(0, \cos(\theta_j^{\text{wind}} - \beta_{j\to i})\big), \qquad (2)$$

where θ_j^{wind} is the direction the past six hours of wind at sender j blow toward and $\beta_{j\to i}$ is the bearing from j to i, following PM2.5-GNN [4]. Two residual diffusion-convolution layers [15] use forward and backward random-walk transitions of $\tilde{A}$ and an adaptive adjacency $\mathrm{softmax}(\mathrm{ReLU}(E_1 E_2^\top))$ learned from node embeddings [16], each with diffusion order $K = 2$:

$$H' = H + \mathrm{ReLU}\left(\left[H;\ \{P^k H\}_{P\in\mathcal{P},\,k=1..K}\right] W\right). \qquad (3)$$

Decoder and loss. A two-layer MLP maps each node state to all 48 leads. In the CAMS-informed setting the latest published CAMS forecast for each lead is added through a per-lead input. Training minimises a masked Huber loss on observed targets only, plus the gating penalty.

V. Experimental Protocol

Splits (UTC). Train 2018–2021, validation 2022, test 2023–2024 (two winters, two stubble seasons and two Diwalis), an extra PM-only test from April to September 2025, and a prospective winter 2026–27 test from live data. The test splits are hashed in a lock file and can only be read by one evaluation script after the model is frozen; tuning, model selection and early stopping use validation only. A second rolling-origin fold (train to mid-2022, validation to mid-2023, test mid-2023 to 2024) checks robustness.

Latency rules. At issue time t_0, station observations are usable up to $t_0 - 1$ h, fire detections from acquisition plus three hours, and a CAMS run R only from $R + 10$ h. Past CAMS inputs therefore use short-lead forecasts of the newest usable run rather than analyses. Normalisation statistics come from the training years. Unit tests perturb every value after t_0 and check that the inputs do not change.

Settings. *S-A* (primary) uses past data only. *S-B* adds the latest published CAMS forecast at matched leads, framing the model as CAMS post-processing. *S-C* adds true future meteorology and is reported only as an upper bound, never as a forecast.

Baselines. Persistence, same hour yesterday, climatology (station × month × hour), raw CAMS at matched lead, CAMS with per-station linear bias correction, LightGBM on lags and meteorology (S-A) and as model output statistics on CAMS (S-B), LSTM, CNN-LSTM, a ConvLSTM baseline, PM2.5-GNN with past-only meteorology [4], and Graph WaveNet [16]. All neural models share the same data loader, loss and training schedule.

Training. AdamW with cosine decay, batch size 32, up to 60 epochs with early stopping on validation RMSE (patience 8), mixed precision and gradient clipping. Hyperparameters are tuned on validation with at most 20 trials per rung. Final models are trained with five seeds. Hardware: [TO FILL: GPU, from compute.md].

TABLE II
Test results, setting S-A (station level). Generated from run artifacts.

Model	+1 h RMSE	+1 h Skill	+24 h RMSE	+24 h Skill	+48 h RMSE	+48 h Skill	DM p
Persistence	–	–	–	–	–	–	–
Climatology	–	–	–	–	–	–	–
LightGBM (S-A)	–	–	–	–	–	–	–
LSTM	–	–	–	–	–	–	–
PM2.5-GNN (past met)	–	–	–	–	–	–	–
Graph WaveNet	–	–	–	–	–	–	–
MW-AeroFusion-Net	–	–	–	–	–	–	–

[TO FILL: generated by make_tables.py –final after final_eval.py]

TABLE III
Test results, setting S-B (with CAMS forecast). Generated from run artifacts.

Model	+1 h RMSE	+1 h Skill vs CAMS	+24 h RMSE	+24 h Skill vs CAMS	+48 h RMSE	+48 h Skill vs CAMS
CAMS raw	–	–	–	–	–	–
CAMS + bias correction	–	–	–	–	–	–
LightGBM MOS (S-B)	–	–	–	–	–	–
MW-AeroFusion-Net (S-B)	–	–	–	–	–	–

[TO FILL: generated by make_tables.py –final after final_eval.py]

Metrics. RMSE, MAE, R^2, bias, and skill $1 - \mathrm{RMSE}/\mathrm{RMSE}_{\text{ref}}$ against persistence and CAMS at +1, +24 and +48 h; breakdowns by season, Indo-Gangetic Plain versus rest, and city; hit rate, false-alarm rate and critical success index for $\mathrm{PM}_{2.5} > 250\,\mu\mathrm{g/m}^3$; AQI-category accuracy and macro-F1. Differences are tested with the Diebold–Mariano test with the HLN correction [38] and a 24 h moving-block bootstrap.

VI. Results

Main comparison. Table II reports test-set errors at the three leads in the primary setting S-A, and Table III in the CAMS-informed setting S-B, as mean ± standard deviation over five seeds. [TO FILL: two or three sentences comparing MW-AeroFusion-Net with persistence, LightGBM and CAMS with bias correction at each lead, quoting only the generated tables and the Diebold–Mariano p-values].

Component ladder and ablations. Table IV lists the validation error of rungs M0–M6, each adding one component, and Table V the leave-one-out ablations of the full model on the test set. [TO FILL: which components reduced error, which did not; a component that did not help is reported here, not removed].

Seasons, regions and episodes. [TO FILL: season (DJF, MAM, JJAS, ON) and IGP-versus-rest breakdown; CSI for $\mathrm{PM}_{2.5}$ above the severe threshold; AQI-class macro-F1; from the generated tables].

Case studies. Fig. 2 shows Delhi during the stubble-burning seasons of November 2023 and 2024 and the two Diwali periods. [TO FILL: one or two sentences on what the forecasts got right and wrong during these events].

S-C upper bound and robustness. With true future meteorology (S-C), [TO FILL: gap between S-A and S-C, labelled as an upper bound]. On the rolling-origin fold, [TO FILL: whether the ranking of models holds]. On the PM-only 2025 check, [TO FILL:

TABLE IV
VALIDATION RESULTS (S-A, SEED 0): BASELINES AND LADDER RUNGS M0–M6. GENERATED FROM RUN ARTIFACTS; RUNG ROWS APPEAR ONCE THE LADDER HAS RUN.

Model	+1 h			+24 h			+48 h		
	RMSE	MAE	Skill	RMSE	MAE	Skill	RMSE	MAE	Skill
climatology	45.4	29.3	-0.908	44.5	28.6	-0.134	44.5	28.6	0.022
persistence	23.8	13.0	0.000	39.3	22.6	0.000	45.5	26.8	0.000
same_hour_yesterday	38.6	22.4	-0.624	44.8	26.3	-0.141	47.6	28.5	-0.047

TABLE V
LEAVE-ONE-OUT ABLATIONS OF MW-AEROFUSION-NET (TEST, S-A, RMSE, FIVE SEEDS).

Variant	+1 h	+24 h	+48 h
Full model (M6)	–	–	–
− wavelet bands	–	–	–
− graph	–	–	–
fixed graph only	–	–	–
− ConvLSTM context	–	–	–
Transformer → GRU	–	–	–
− gating	–	–	–

[TO FILL: fill from ablation runs via make_tables.py –final]

result]. On the prospective winter 2026–27 test, [TO FILL: result, or state that it was not available].

Compute. MW-AeroFusion-Net has [TO FILL: parameter count] parameters; training took [TO FILL: time per epoch and total GPU-hours]; ONNX inference on a CPU takes [TO FILL: latency] per forecast (from compute.md).

VII. ATTRIBUTION VALIDATION

We test the gate weights with four checks, each reported as passed or failed (Table VI). (a) *Faithfulness:* masking the top-k gated groups should raise the error more than masking random groups, and compares with the top-k groups by Integrated Gradients [33]. (b) *Sanity:* randomising the weights layer by layer should change the attributions [32]. (c) *Stability:* group rankings should agree across seeds. (d) *Plausibility:* for Indo-Gangetic Plain stations in October–November, the fire-group weight should rise with upwind FIRMS counts; in March–May the dust-group weight should rise with CAMS dust optical depth; and the traffic-proxy weight should fall during the 2020 lockdown relative to the same weeks of 2019. We also compare group rankings with TreeSHAP on LightGBM [34]. [TO FILL: summary of which checks passed and which failed].

VIII. LIMITATIONS

ERA5 reanalysis stands in for operational meteorological analyses during training; the live system uses forecast meteorology, so served skill may differ from benchmark skill. CPCB coverage after 2024 is sparse, so the 2025 check uses $PM_{2.5}$ and PM_{10} only, and live station data arrive with a delay of about two days, which limits real-time use of the past-only setting. Five study cities have no long-running station and are scored only through their neighbours. Traffic is represented by proxies (NO_2, CO and the calendar), since no traffic data exist at this scale. The gate weights are learned associations; they passed [TO FILL: the checks that passed] but are not a causal

[TO FILL: figure from make_figures.py –cases]

Fig. 2. Observed and forecast city-mean $PM_{2.5}$ in Delhi at +24 h during stubble-burning and Diwali periods of the test years.

TABLE VI
ATTRIBUTION CHECKS FOR THE GATING LAYER (TEST, S-A).

Check	Statistic	Outcome
(a) Deletion: gated vs random	–	–
(a) Deletion: gated vs Integrated Gradients	–	–
(b) Weight randomisation	–	–
(c) Seed stability (rank corr.)	–	–
(d) Fire weight vs upwind fires, ON	–	–
(d) Dust weight vs CAMS dust AOD, MAM	–	–
(d) Traffic proxy, lockdown vs 2019	–	–
Agreement with LightGBM TreeSHAP	–	–

[TO FILL: from explain.py outputs]

source apportionment. [TO FILL: any further limitation found in the results].

IX. CONCLUSION

We built a leakage-safe benchmark for station-level $PM_{2.5}$ forecasting over 28 Indian cities and used it to evaluate MW-AeroFusion-Net component by component against CAMS forecasts and strong statistical and neural baselines. [TO FILL: main finding in one or two sentences, from the results only]. [TO FILL: what the ablations show about which components matter]. Because every input respects its publication latency and the test years were opened once, these numbers are meant to reflect what an operational forecaster could expect. Future work will [TO FILL: next steps, for example a near-real-time station feed and longer prospective evaluation].

ACKNOWLEDGMENT

Data: CPCB CAAQM via the XKDR Forum India Air Quality API (CC BY 4.0); ERA5 and CAMS from the Copernicus Climate Change and Atmosphere Monitoring Services (CC BY 4.0); FIRMS fire data from NASA. The results contain modified Copernicus information; neither the European Commission nor ECMWF is responsible for any use of it. AI tools were used to assist with code and drafting; all results are generated from run artifacts and were checked by the authors.

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